Arun V

+91 98414 45400 | arinvijey@gmail.com | linkedin.com/in/arun-vijey/

Education

Chennai Institute of Technology — Chennai, India
Bachelor of Engineering - Mechanical Engineering (CGPA - 8.62) — *Sep 2021 – April 2025*

Experience

Project Associate — Oct 2025 – Present
IIT Madras - Ballistics and High Speed Flow Laboratory — *Chennai, India*
- Performed CFD & structural simulations using ANSYS Fluent and LS - DYNA for project research.
- Completed end-to-end CFD and FEA workflows, including Geometry preparation, Meshing, Solver setup, Post-processing, Validation and delivered technical insights through documentation

UAV Intern — Dec 2024 – Feb 2025
Xagrotor Tek Private Limited — *Chennai, India*
- Contributed to composite manufacturing processes for lightweight UAV components.
- Operated laser cutting and CNC machines to prototyping VTOL structures.
- Assisted in agricultural drone DGCA type certification and in drafting designs for a 3.8 m fixed-wing VTOL drone.

Project Intern — Mar 2023 – Sep 2023
Rane Madras Ltd (RML) — *Chennai, India*
- Developed a Python - based automation program for bend prediction, improving process efficiency.
- Modified an Auto Rack Straightener machine to enable fully autonomous operation.
- Implemented communication between PC and machine using LabVIEW and PLC programming.

Mechanical Intern — Aug 2022 – Sep 2022
National Institute of Wind Energy — *Kayathar, India*
- Conducted wind farm visits, observed turbine maintenance, and installed a 30 m wind mast for data collection.
- Gained insights into wind farm planning and authored a detailed review and delivered on-campus presentations.

Projects

Wear Behavior of YSZ-Reinforced AMC via Ultrasonic Squeeze Casting — Dec 2024 – Mar 2025
- Developed aluminium matrix composites using aluminium scrap and yttria-stabilized zirconia reinforcement.
- Used ultrasonic-assisted squeeze casting to improve material properties and conducted wear testing to evaluate performance in different conditions.

Propeller Noise Reduction and Optimization | *CAD, CFD, Aeroacoustics* — May 2024 – Sep 2024
- Designed unconventional propellers to evaluate lift and noise performance against traditional counterparts.
- Tested 10+ design iterations using Ansys Fluent to determine optimal performance and visualized the data in CFD Post to show improvements in noise reduction by 30%

Achievements

- Secured AIR 7, TN Rank 1, and Overall Rank 1 in Technical Presentation (Micro Class Aircraft). (Drone Development Challenge 2023 SAE Southern Section)
- Secured 4th place in CFD and 1st place in Design Report (Micro Class, DDC 2025 SAE Southern Section).
- Secured 3rd place in the National Level CFD Competition using Ansys Fluent. (NLCCFD SAE Southern Section)

Technical Skills

CAD Design: CATIA, SolidWorks
Analysis: Ansys Fluent, Static Structural, Explicit Dynamics, LS-DYNA
Programming Languages: Python, C, Java, HTML/CSS

Certifications

- 3DEXPERIENCE ASSOCIATE (Dassault Systèmes) for Mechanical, Additive Manufacturing, Electrical Design, Sustainability and 3D Mold Creator